Design of a 24-Hour Digital Clock Using Flip-Flops

Electronic Devices and Circuits Laboratory

R Rahul
B.Tech Electrical Engineering, IIT Hyderabad

October 16, 2026

Abstract

This report presents the design concept of a 24-hour digital clock based on sequential digital logic. The clock maintains seconds from 00 to 59, minutes from 00 to 59, and hours from 00 to 23. The design uses JK and D flip-flops, synchronous and divide-by-N counters, combinational decoding logic, and reset and rollover logic. The report summarizes the system architecture, operating principles, design flow, suggested verification cases, challenges.

Contents

1 Introduction **2**

2 Objectives **2**

3 System Specifications **2**

4 Block-Level Architecture **2**

5 Main Building Blocks **3**
5.1 Flip-Flops . . . 3
5.2 Counters . . . 3
5.3 Carry and Rollover Logic . . . 4
5.4 Reset Logic . . . 4
5.5 Display Decoding . . . 4

6 Design Flow **4**

7 Verification and Test Plan **4**

8 Challenges and Debugging Considerations **5**

9 Applications **5**

10 Conclusion **6**

1 Introduction

A digital clock is a sequential circuit that updates its displayed time in response to periodic clock pulses. A 24-hour clock represents time from 00:00:00 through 23:59:59 and then returns to 00:00:00. This project demonstrates how flip-flops and counter logic can be combined to build a modular timekeeping circuit.

The supplied project description identifies this as an Electronic Devices and Circuits (EDC) Lab project and describes the use of JK and D flip-flops, synchronous counters, divide-by-N counters, combinational logic, and rollover detection.

2 Objectives

- Design a sequential circuit that counts seconds, minutes, and hours in 24-hour format.
- Understand the role of JK and D flip-flops as state-storage elements.
- Implement modular counters and carry propagation between time units.
- Design rollover and reset logic for transitions at 59 seconds, 59 minutes, and 23 hours.
- Analyze timing behavior and verify stable state transitions.
- Decode counter states for display.

3 System Specifications

The functional ranges stated in the project README are summarized in Table 1.

Table 1: Functional specifications

Parameter	**Specification**
Clock format	24-hour
Seconds range	00–59
Minutes range	00–59
Hours range	00–23
Sequential elements	JK and D flip-flops
Counter types	Synchronous and divide-by-N
Control logic	Reset, carry, and rollover
Output stage	Display decoding logic

We didn't specify the clock input frequency, display technology, component part numbers, measured timing results, or hardware test results. These details should be added only after they have been established from the circuit files or measurements.

4 Block-Level Architecture

The system is organized into counter stages connected by carry signals. The seconds counter advances first. When it rolls over from 59 to 00, it generates a carry to increment

the minutes counter. When the minutes counter rolls over from 59 to 00, it generates a carry to increment the hours counter. The hours counter rolls over from 23 to 00.

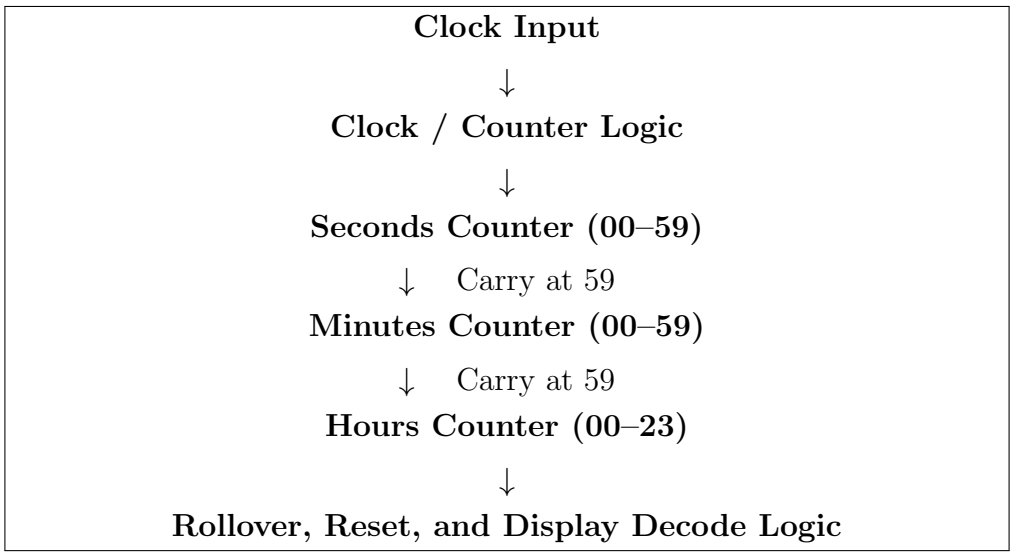


Figure 1: Conceptual block diagram of the 24-hour clock.

5 Main Building Blocks

5.1 Flip-Flops

JK and D flip-flops are identified in the project description as the basic storage elements. A flip-flop stores one bit of state. Multiple flip-flops are combined to represent counter states and control the next state on clock edges.

A D flip-flop transfers its input D to output Q on the active clock edge, subject to its triggering and reset behavior. A JK flip-flop can hold, set, reset, or toggle depending on its J and K inputs.

5.2 Counters

The clock uses counters with different terminal counts:

- Seconds: counts from 00 through 59.
- Minutes: counts from 00 through 59.
- Hours: counts from 00 through 23.

The counter implementation must ensure that states outside these ranges are not entered during normal operation.

5.3 Carry and Rollover Logic

Carry signals connect the counter stages. The expected rollover behavior is:

$$\begin{aligned} 59 \text{ seconds} &\rightarrow 00 \text{ seconds, increment minutes,} \\ 59 \text{ minutes} &\rightarrow 00 \text{ minutes, increment hours,} \\ 23 \text{ hours} &\rightarrow 00 \text{ hours.} \end{aligned}$$

These transitions require combinational detection of terminal states and suitable synchronous next-state or reset logic. The precise implementation depends on the circuit.

5.4 Reset Logic

Reset initializes the clock to a known state and can help recover from invalid counter states. The reset polarity, synchronous or asynchronous behavior, and initial time are not specified in the README and should be documented from the actual design.

5.5 Display Decoding

The project includes display/decode logic as the final stage. The decoder translates the internal counter state into a representation suitable for the chosen display. The README does not identify the display hardware or decoder IC, so no particular display implementation is assumed in this report.

6 Design Flow

1. Provide a periodic clock input.
2. Generate the required counting pulses using clock or divide-by-N counter logic.
3. Update the seconds counter on each applicable pulse.
4. Generate a carry when seconds roll over from 59 to 00.
5. Update the minutes counter when the seconds carry is asserted.
6. Generate a carry when minutes roll over from 59 to 00.
7. Update the hours counter when the minutes carry is asserted.
8. Roll hours over from 23 to 00.
9. Apply reset and decode the states for display.

7 Verification and Test Plan

The timing analysis, reset behavior, rollover, and stable state transitions, but it does not provide numerical simulation results or measured hardware data. The following cases are proposed verification tests, not claims that testing has already been completed.

Table 2: Suggested functional test cases

Test	**Initial State**	**Expected Result**
Seconds increment	00:00:08	00:00:09
Seconds rollover	00:00:59	00:01:00
Minutes rollover	00:59:59	01:00:00
Hour increment	07:24:59	07:25:00
Day rollover	23:59:59	00:00:00
Reset operation	Any valid state	Documented reset state

During simulation or hardware testing, observe the counter outputs, carry signals, and reset response. Record the clock frequency and period, timing assumptions, waveforms, and any unexpected transitions. If a display is present, confirm that its decoded output matches the internal counter state.

8 Challenges and Debugging Considerations

The following potential challenges:

- Timing mismatches between sequential blocks.
- Incorrect rollover conditions.
- Reset behavior that does not initialize all counters consistently.
- Unstable or incorrect state transitions.
- Invalid transitions during counter rollover.

A useful debugging procedure is to inspect the counter state and carry signal immediately before and after each terminal count. For example, at 00:00:59, the seconds counter should return to 00 and the minutes counter should increment exactly once. If the minutes counter increments more than once or not at all, inspect the terminal-count decode, carry pulse width, and clocking relationship between counter stages.

This is a recommended debugging method; the supplied README does not identify a specific fault that was actually encountered or fixed. Any report of a completed fix should describe the fault and correction observed in the real simulation or hardware.

9 Applications

The design concepts are applicable to:

- Digital clocks and timers.
- Event counters and frequency dividers.
- Digital control systems.
- Sequential logic systems.
- FPGA and RTL-based digital designs.

10 Conclusion

The 24-hour digital clock project demonstrates the use of flip-flops, synchronous counters, divide-by-N counters, combinational decoding, reset logic, and rollover detection to represent time in hours, minutes, and seconds. The key functional requirement is correct progression through 00–59 seconds, 00–59 minutes, and 00–23 hours, including the transition from 23:59:59 to 00:00:00.